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Abundance of flagellate and ciliate protozoa in some agriculture field of Madhepura district

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Abstract- Physicochemical analysis of soil, abundance of active flagellate and ciliate protozoa was studied in Maize, Wheat and Paddy fields of Madhepura district. Highest pH was recorded from Maize field of Gwalpada where abundance of flagellate and ciliate protozoa in rhizosphere of crop observed as 22 ind/gm and 28 ind/gm soil respectively. Lowest abundance of flagellate and ciliate protozoa was observed in paddy field (flagellate-11 ind/gm, ciliate-12 ind/gm) where pH value was 5.8. Further it was observed that the surface soil contain higher abundance of protozoa in comparison to deep soil. At a depth of 15-30cm abundance of protozoa is lowest.

Keywords: pH, Abundance, Flagellate protozoa, Ciliate protozoa

INTRODUCTION

A large number of microbes in soil play essential role for food web security. Protozoa are the higher trophic members of the soil food chain than bacteria. They feed on bacteria and release nutrient which is subsequently taken up by growing crops. Soil protozoa represent a very important microbial group within the soil community, with high abundance and ecological role in respect to nutrient cycling.[1] Soil protozoa regulate the size and composition of soil bacterial community.[2] They enhance the growth of earthworms and crops.[3] Ciliate protozoa play a major role in soil by regulating the growth of bacteria and other smaller protists.[4] Many ciliate protozoa are considered as markers of environmental stress in the soil ecosystem.[5] Abundance of ciliate protozoa in soil had a strong correlation with physico-chemical structure of soil. Many studies has carried out to assess the effect of pollutants such as trace elements and pesticides on soil protozoa particularly on ciliate protozoa has shown that they are useful bio indicators in agriculture systems.[4] Soil protozoa stimulate growth of PGPR (plant growth promoting rhizobacteria). These bacteria are able to fix atmospheric nitrogen, produce plant growth hormones and inhibit the growth of pathogenic bacteria.

MATERIAL & METHODS

Soil samples were collected from Maize, Wheat and Paddy fields of 5 villages namely Gwalpada, Haripur, Ghelar, Bhelwa and Shankarpur in Madhepura district. List of villages and soil sample codes are mentioned in Table 1. Soil samples were collected from two different depts. (0-15 cm and 15-30 cm).

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Table 1- List of villages and soil sample codes.

Villages	Crop	Soil sample code
Gwalpada	Maize	GWA
Haripur	Wheat	HAR
Ghelar	Wheat	GHE
Bhelwa	Paddy	BHE
Shankarpur	Maize	SHA

Abundance of active Flagellets: One gram soil was dissolved in 5ml water in a petridish. Suspension was mixed and allowed to stand for 30 seconds. Suspension was stained with 0.015ml silver stain. Suspension was taken out with the help of pipette and transferred in Haemocytometer chamber and observed under microscope. Number of flagellate Protozoa were counted. Abundance was calculated by the formula:

$$Abundance\ of\ flagellete\ Protozoa/gm = \frac{x\ X\ 10}{0.015}$$

Where,

x = No. of flagellate in 0.015ml suspension.

Abundance of active Cilliates: Each soil sample was homogenized by through mixing in a glass bowel. 5gm homogenized soil sample was taken in a petridish and dried at room temperature for 5 days. Dried soil was passed through a 4ml sieve. In sieved soil rain water was added to produce a slurry. Surry was filtered through a filter of 0.2µm pore size. Filtrate was stained by silver staining. One drop (0.1ml) of stained sample was transferred in Sedgewick-Rafter chamber and number of ciliated Protozoa were counted under microscope. It was repeated three times. Abundance of ciliated protozoa was calculated by the formula:

$$Abundance\ of\ ciliated\ Protozoa/gm = \frac{x\ X\ (V1 = V2)}{50\ X\ W}$$

Where,

x = No. of ciliated Protozoa as counted in Sedgewick-Rafter chamber

V1 = Volume of rain water added in 5gm soil

V2 = Volume of water determined in 5gm soil sample

W = Weight of soil sample (5gm)

Physiochemical parameters of soil samples were examined for pH, E.C., Carbon, Nitrogen, P and K.

RESULT

Soil samples were collected from Maize, Wheat and Paddy fields of 5 villages under Madhepura district. Soil samples were collected from two different depth i.e. from 0-15cm and 15-30cm as well as from rhizosphere of crops. Physicochemical parameters of soil were examined. pH ranged in between 5.8 to 8.2. Lowest pH was recorded from soil sample of paddy field. Electrical conductivity ranged in between 0.24 to 0.39dcm^{-1}. The result of pH and E.C. is shown in Fig. 1 and 2 respectively.

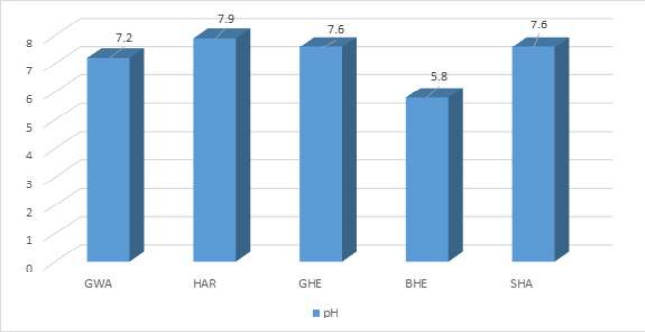


Fig. 1- pH value of different soil samples

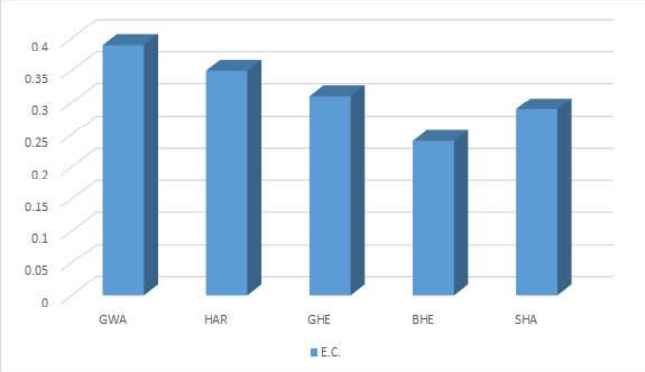


Fig. 2- Electrical conductivity of different soil samples

Organic carbon was highest in soil sample code BHE (0.62%) and lowest in soil sample code HAR (0.54%). Nitrogen was highest in soil sample BHE (61.2%) and lowest in soil sample SHA (55.6%). Phosphorus was highest in soil sample BHE (7.2mg/g) and lowest in soil sample GWA (5.9mg/g). Potassium was highest in soil sample GWA (210.5kg/h) and lowest in soil sample GHE (190.6kg/h). The result is mentioned in Table 2.

Table 2- Physicochemical parameters of soil samples

Soil sample	OC %	N %	P mg/g	K kg/h
GWA	0.58	58.6	5.9	210.5
HAR	0.53	57.4	6.7	205.3
GHE	0.56	59.3	6.8	190.6
BHE	0.62	61.2	7.2	195.5
SHA	0.54	55.6	6.4	203.7

Abundance of flagellate and ciliate protozoa were highest in rhizospheric soil and lowest in 15-30cm deep soil. It was also observed that the abundance of protozoa was low at lower pH and high in alkaline pH. In rhizospheric soil, lowest abundance of protozoa was recorded in soil sample BHE collected from rhizosphere of paddy crop (flagellate-11, ciliate-12) and highest in rhizospheric soil GWA collected from rhizosphere of Maize crop (flagellate-22, ciliate-28). The result is shown in Fig. 3 to 7.

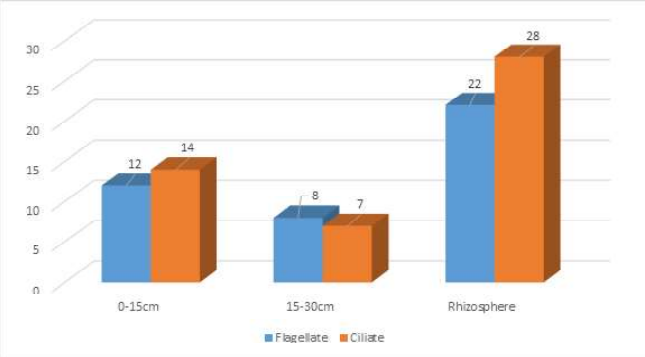


Fig. 3- Abundance of flagellate and ciliate protozoa per gram soil of Gwalpada

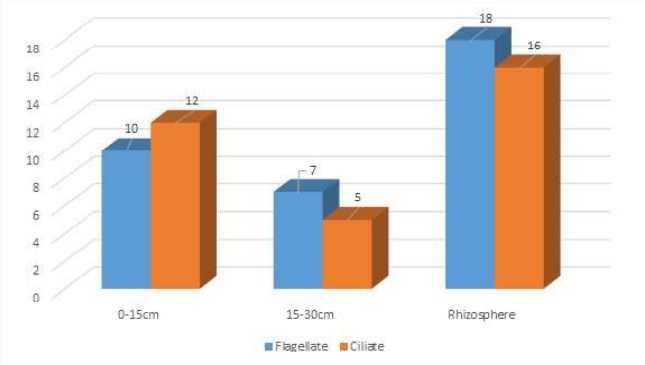


Fig. 4- Abundance of flagellate and ciliate protozoa per gram soil of Haripur

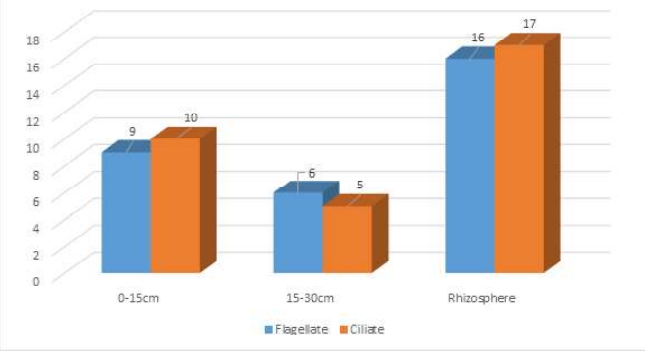


Fig. 5- Abundance of flagellate and ciliate protozoa per gram soil of Ghelar

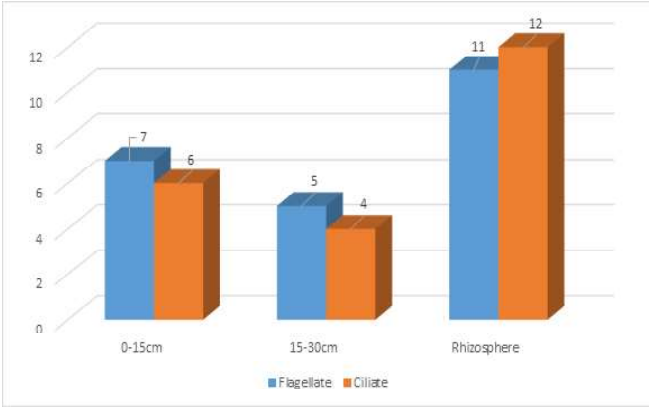


Fig. 6- Abundance of flagellate and ciliate protozoa per gram soil of Bhelwa

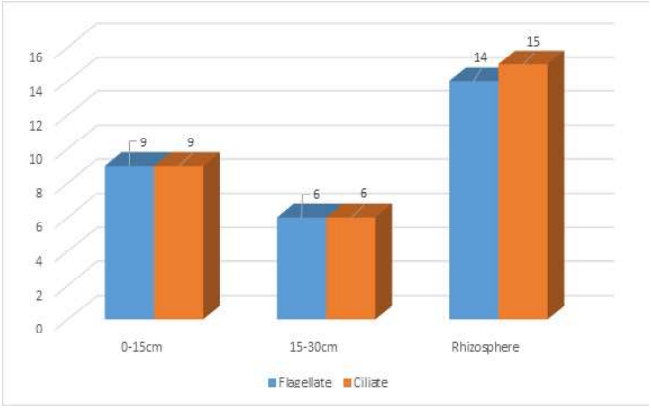


Fig. 7- Abundance of flagellate and ciliate protozoa per gram soil of Shankarpur

DISCUSSION

Several studies have been conducted on physicochemical analysis of soil including correlation with microbial diversity. Out of physicochemical parameters pH is very important parameters as it determines the availability of nutrients and physical condition which control the microbial activity.[6] Electrical conductivity and pH frequently affect distribution and abundance of protozoa particularly ciliate protozoa.[7] In the present study, it was observed that the abundance of ciliate protozoa was lower in low pH while high in alkaline pH. Nitrogen content of soil also affect abundance of protozoa. Low amount of nitrogen content decreases the abundance of protozoa.[8] In present study, it was also observed that the abundance of both flagellate and ciliate protozoa were high where nitrogen content of soil was high. Schwarz and Frenzel-(2003)[9] reported that the number of protozoa decreases in flooded paddy field. In our study, it was also observed that abundance of flagellate and ciliate protozoa was lower in

paddy field in comparison to abundance of both type of protozoa in soil samples of Maize and Wheat field.

CONCLUSION

In the present study, abundance of flagellate and ciliate protozoa was studied in different crop fields of Madhpura district. Physicochemical parameters of soil were also studied. It was observed that the abundance of protozoa was highest in rhizospheric soil. In rhizosphere of paddy field abundance of flagellate protozoa was recorded as 11 ind./gm soil and abundance of ciliate protozoa as 12 ind.gm soil. Abundance of protozoa was lowest in paddy field where pH was also lowest (5.8). Highest abundance of protozoa was recorded in rhizospheric soil of Maize crop from Gwalpada (flagellate-22, ciliate-28) where pH was also highest (8.2). Abundance of both flagellate and ciliate protozoa in rhizospheric soil of other Maize crop from Shankarpur was recorded as 14 ind./gm soil and 15 ind./ gm soil where pH was 7.2. Abundance of flagellate and ciliate protozoa in rhizosphere of Wheat in two fields of Haripur and Ghelar was recorded as 18 ind./gm and 16 ind./gm soil for flagellate and 16 ind./gm, 17 ind./gm soil for ciliate protozoa. Abundance of both group of protozoa was high in upper soil and lower at the depth of 15-30cm.

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